

化学理论 (I) 量子力学

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化学理论I (量子力学)

- 量子力学基础(薛定谔方程, 算符)
- 量子力学的简单体系 (势阱、谐振子、刚性转子、类氢原子)
- 多电子体系基本概念 (自旋、全同粒子、反对称化波函数)
- 近似方法
- 多电子体系 (多电子原子与双原子分子)

化学理论I (对称性与分子光谱)

- 分子光谱理论

分子振动、转动能级及跃迁理论与红外、拉曼光谱。

- 分子对称性与群表示理论

点群、群表示与约化、对称轨道(SALC)、对称性与积分、

- 多原子分子结构

由对称轨道构筑分子轨道、分子轨道理论与配位场/晶体场理论、HMO近似

- 简正振动的对称性分析与分子光谱

- 统计基础与热力学物理量基本概念

从微观粒子入手，描述大量粒子所形成的体系与对应的玻尔兹曼分布。描述宏观的热力学物理量概念，介绍粒子数，熵，吉布斯自由能，内能等多个概念以及所对应的物理量关联与计算方法。

- 热力学基本思想与假设

讲述宏观与微观的概念，通过微观粒子的状态数计算延伸到与之对应的宏观物理量上。前提：等概率假设，不可逆过程，玻尔兹曼熵定义，孤立系统假设，熵第二定律（熵增）。

- 配分函数

利用配分函数来计算热力学物理量，介绍亥姆霍兹能，非相互作用粒子，可区分与不可区分粒子的分配函数，能级介绍与零点能选取，对热力学方程的计算，状态数，平动配分函数，转动配分函数，振动方程，电子配分函数，热力学贡献。

- 熵

理解能均分定则，熵贡献（平动，转动，振动）。

- 热容与内能贡献

- 自旋统计

波函数对称性，费米子与玻色子，粒子交换。

- 化学平衡

掌握化学势平衡，亥姆霍兹能配分方程表达化学势，电子状态数测量，孤立原子基态测量，双原子热解离，原子热电离与撒哈方程，气体表面吸附，Langmuir等温线推导。

- 速率常数

掌握势能面，过渡态理论，动力学同位素效应，过渡态理论中的Eyring方程，阿伦尼斯方程，溶液反应，气相反应，指数前系数。

- 玻尔兹曼分布

- 量子力学与固体物理基础

理解固体物理的发展历程、固体电子结构的研究意义与研究方法；掌握量子力学的基本概念。

- 晶体结构与晶体衍射

掌握晶体的特征，学会描述晶体结构（点阵+基元=晶体结构）；了解三维晶格点阵的对称性、七大晶系和14种布拉菲晶格；理解倒易点阵的概念、推导及其与晶体点阵之间的关系；理解倒易点阵与晶体衍射的关系及Ewald球构建晶体衍射的过程和原理。掌握布里渊区的概念，掌握Bragg定律、Laue方程、Brillouin 边界衍射条件的等价关系。

- 化学键与电子结构

掌握根据化学键(金属键、共价键、离子键、范德华键)的特点理解所形成晶体的电子结构及性质；理解金属、半导体导电率随温度的变化，通过离子模型理解离子晶体电子结构，金属氧化物电子结构的变化规律。掌握半导体的分类及其应用。

- 自由电子模型

基于特鲁德模型的几个基本假设及相关物理概念，利用特鲁德模型定性解释金属中电子对导电和热容的贡献。分析索末菲模型下电子对热容的贡献；深入理解维德曼-弗兰兹定律，以及金属导热、导电的微观机制。分析特鲁德经典自由电子气、索末菲自由电子模型

- 能带理论

能带理论中的三个基本近似和处理问题的思路；对布洛赫定理进行详细的证明和推导，并解释其物理意义；通过对克隆尼克-潘纳模型的推导，深入分析能带产生的本质原因；

介绍近自由电子近似模型的思路和物理图像，通过量子力学微扰论求解能带结构的过程，使其理解掌握禁带产生的原因和机制，并详细阐述一维、三维能带结构的构建思路。

介绍紧束缚模型的求解过程和物理图像以及电子形成能带的物理图像，分析金属、半导体的能带结构

1. *Molecular Quantum Mechanics*, P. W. Atkins and R. S. Friedman, Oxford University Press

2. 物质结构 徐光宪 科学出版社

3. 结构化学 周公度等 北京大学出版社

4. 物理化学 一种分子途径 Donald A. McQuarrie
John D. Simon



双一流建设背景下化学系本科量子力学课程模块建设

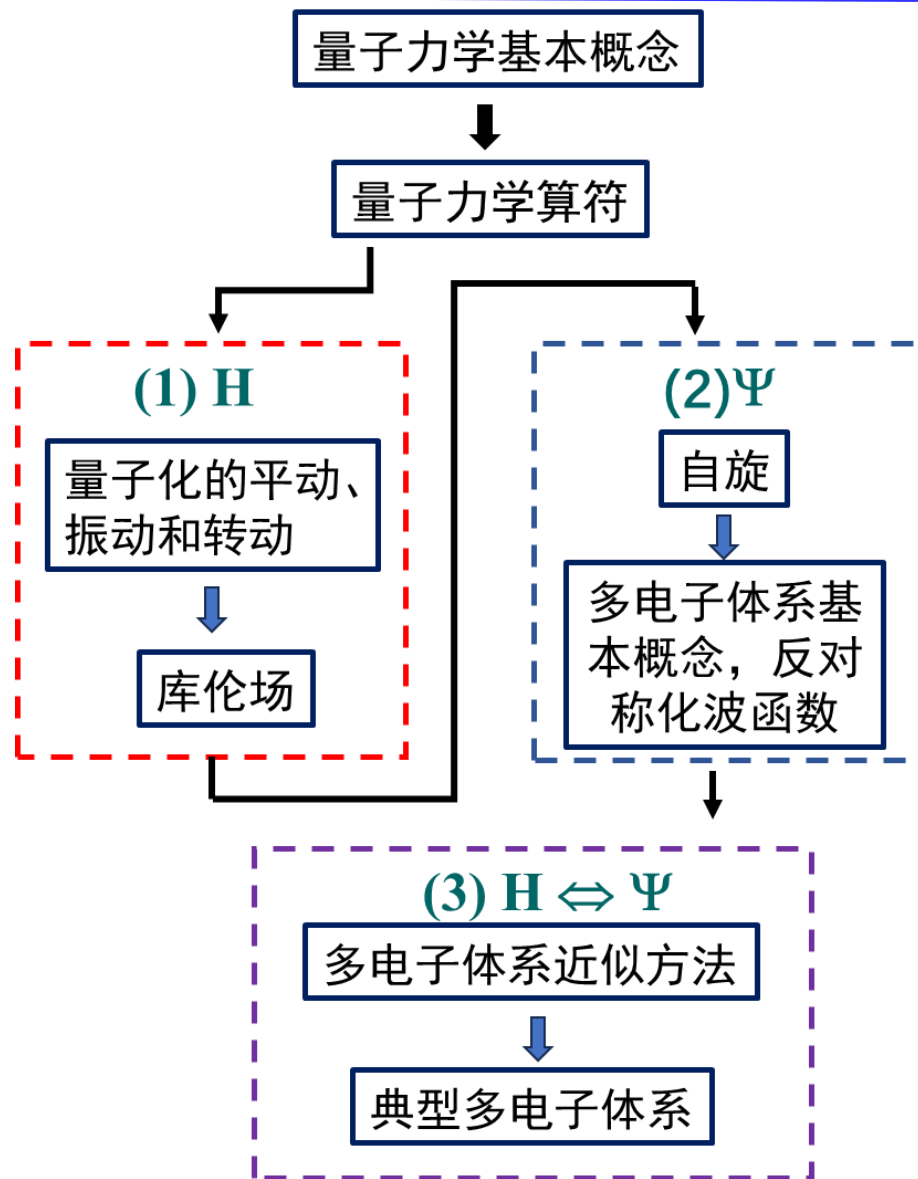
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摘要: 针对目前化学专业本科涉及量子力学内容的教学现状, 整合结构化学和大学普通物理相关教学内容, 厦门大学结构化学课程组开设了化学专业本科量子力学模块课程。本文对该新课程的教学内容与教学方式进行了探讨, 并对该课程的未来发展进行了展望。

量子力学模块知识框架与教学路线



课程要求与成绩

课程要求：

掌握以量子力学处理简单微观体系的方法；以量子力学模型思维解析化学体系，指导后续化学模块的学习。

成绩组成：学伴 + 期中考 + 单元考

Chapter 1. 量子力学基本概念

1. Black-Body Radiation

$$E = nh\nu$$

Planck's constant:

$$h = 6.626 \times 10^{-34} \text{ J}\cdot\text{s}$$

2. The photoelectric effect

A corpuscular theory of light (photons)

$$\varepsilon = h\nu \quad h = \text{Planck's constant}$$

$$p = h/\lambda$$

3. Bohr model



**P. Dirac
(1902-1984)**



**N. Bohr
(1885-1962)**



**W. Heisenberg
(1901-1976)**



**M. Born
(1882-1970)**



**W. Pauli
(1900-1958)**



**L. de Broglie
(1892-1987)**



**M. Planck
(1858-1947)**



**E. Schrödinger
(1887-1961)**

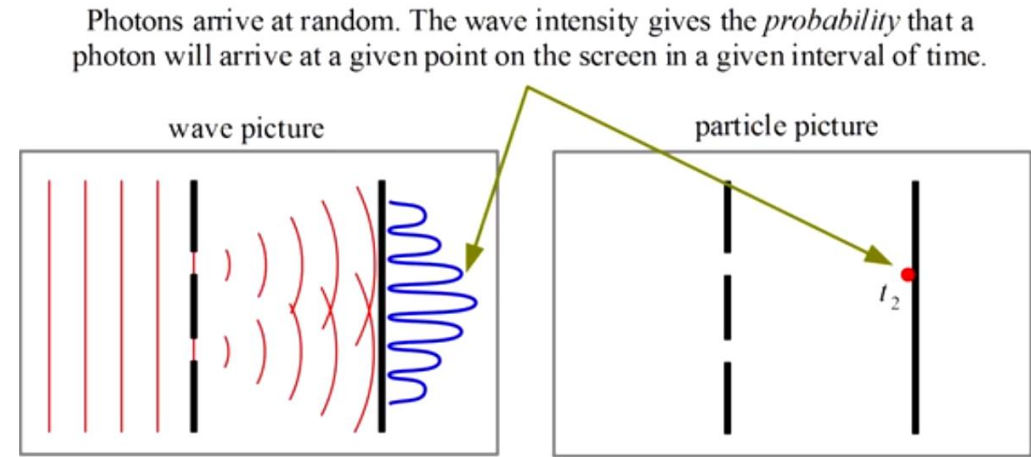
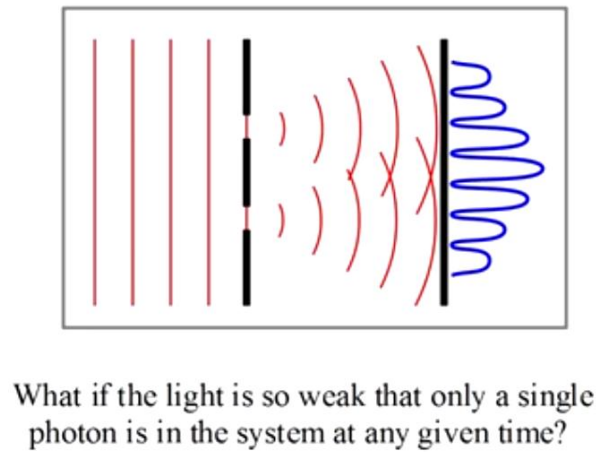


**A. Einstein
(1878-1955)**



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双缝衍射实验



用微弱电子流做双缝衍射实验，让电子几乎是一个一个地通过窄缝，那么时间足够长之后发现：电子在屏上的密度分布不是简单地等于两缝分别开启时的密度分布的和，而是象光波和声波那些经典波一样，出现了两缝相互影响的干涉项。

Wave–particle duality of C_{60} molecules, *Nature* **401**, 680–682 (1999)

Shall we give up determinism?

波-粒子二象性

$$E=h\nu$$

$$p=h/\lambda$$

h = Planck's constant,
 p = particle momentum,
 λ = de Broglie wavelength

- Wave (i.e., light)
 - can be wave-like (diffraction)
 - can be particle-like ($p=h/\lambda$)
- Particles
 - can be wave-like ($\lambda =h/p$)
 - can be particle-like (classical)

波-粒子二象性：每个粒子不仅可以用粒子，也可以用波来描述。它表达了经典概念“粒子”或“波”无法完全描述量子尺度物体的行为。因为粒子有时表现得像波或表现出波的特性，所以很难精确地测量位置和速度。

德布罗意波长

Particles	Mass (g)	Speed (m/s)	λ (m)
Slow electron	9×10^{-28}	1.0	7×10^{-4}
Fast electron	9×10^{-28}	5.9×10^6	1×10^{-10}
Alpha particle	6.6×10^{-24}	1.5×10^7	7×10^{-15}
One-gram mass	1.0	0.01	7×10^{-29}
Baseball	142	25.0	2×10^{-34}
Earth	6.0×10^{27}	3.0×10^4	4×10^{-63}

物质波的本质

光的宏观表现是波动性的，粒子性是其微观特征；

实物粒子的宏观表现是粒子性的，波动性是它的微观特征。

物质波的波动特性不能用经典物理的波给予理解，微观粒子的波动性导致粒子在空间和时间上的概率性分布行为。这意味着粒子出现在空间某一特定位置具有不确定性。这使得测量粒子的空间位置时出现了位置不确定度。

→ 具有波动性的微粒在空间的运动没有确定的轨迹，只有与其波强度大小成正比的几率分布规律。

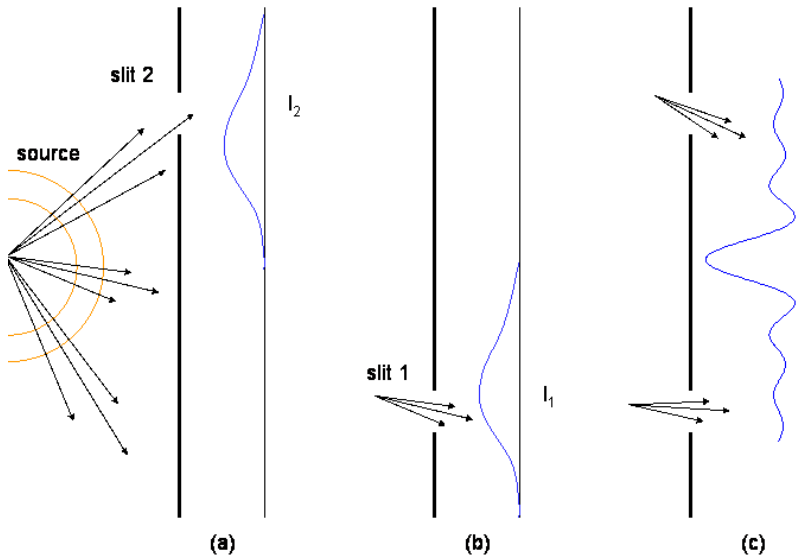
1.3 量子力学系统用其波函数来描述

微观粒子的运动不能用轨道运动的方式来描述，我们无法确知微观粒子从一个状态变化到另一个状态的中间过程。量子理论无法描述粒子状态的中间过程，只能描述微观粒子的状态，由波函数表达。

波函数是系统的数学映射：它本身包含了系统上任何实验都可能测量的所有信息。我们必须对波函数做数学运算来提取这些信息。要获得的最简单的信息是在特定位置找到系统的概率；与波函数的模的平方成正比。

对双缝衍射实验的再认识

电子通过第一个缝（另一个缝关闭）的概率为 P_1 ，电子通过第二个缝（另一个缝关闭）的概率为 P_2 。两个狭缝都是开放的，结果显示在(c)中，观察到一个干涉模式。



$$P = |\Psi|^2$$

$$P_1 = |\Psi_1|^2$$

$$P_2 = |\Psi_2|^2$$

The wavefunction

$$\Psi = \Psi_1 + \Psi_2$$

The superposition principle gives the resultant wavefunction for both slots being opened. Until a measurement is made, the system is "in" all possible states. Here, the possible states are the electron going through slot one (Ψ_1) and the electron going through slot two (Ψ_2). The corresponding intensity is,

$$P = | \Psi_1 + \Psi_2 |^2$$

$$P = | \Psi_1 + \Psi_2 |^2 = P_1 + P_2 + \text{Re} (\Psi_1^* \Psi_2) + \text{Re} (\Psi_1 \Psi_2^*)$$

The final term is called the "interference term." This results in the oscillation pattern in (c).

量子态的叠加原理由此解释了量子干涉现象。当两个狭缝同时开启时，系统的量子描述是两个狭缝单独开启时状态的叠加（即 $\Psi = \Psi_1 + \Psi_2$ ），正是这种叠加态导致了干涉现象。这一规律在确定电子具体选择哪条路径时失效

1.4 The probability

Ψ is a probability amplitude wave! $|\Psi|^2$ tells us the probability of finding the particle at a given place at a given time. Leads to indeterminacy in the fundamental laws of nature.

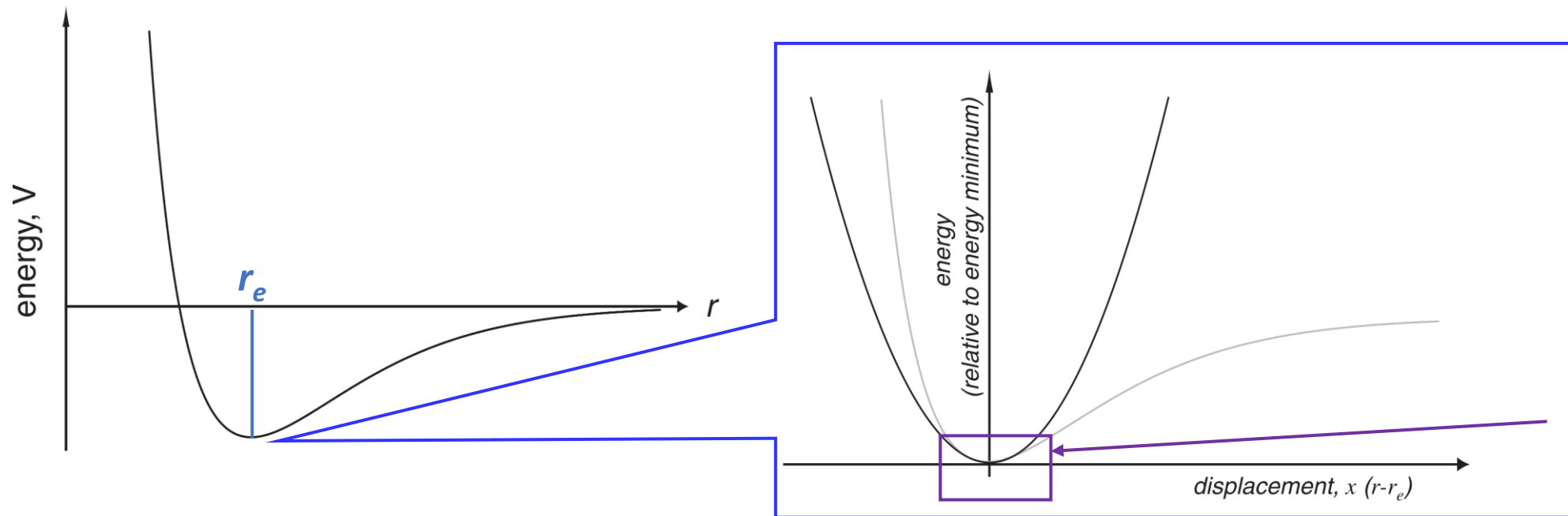
$$\psi^*(r,t)\psi(r,t)dxdydz$$

The probability that the particle lies in the volume element $dxdydz$, located at r , at time t . It should be **normalized**.

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \psi^*(r,t)\psi(r,t)dxdydz = 1$$

$$c \int_{-\infty}^{+\infty} |\psi|^2 d\tau = 1 \quad c = \frac{1}{\int_{-\infty}^{+\infty} |\psi|^2 d\tau}$$

Example: Potential energy curve of a diatomic molecule



- Restoring force acting on the molecule : $-dV/dr$
- At the equilibrium separation r_e , $-dV/dr = 0$
- The simplest assumption: restoring force $\propto -(r-r_e) = -kx$ (let $x = r-r_e$)
 $k = \text{force constant}$ (in N m^{-1})
- The potential energy can thus be given by: $V(x) = -\int -kx dx = kx^2/2 + \text{constant}$

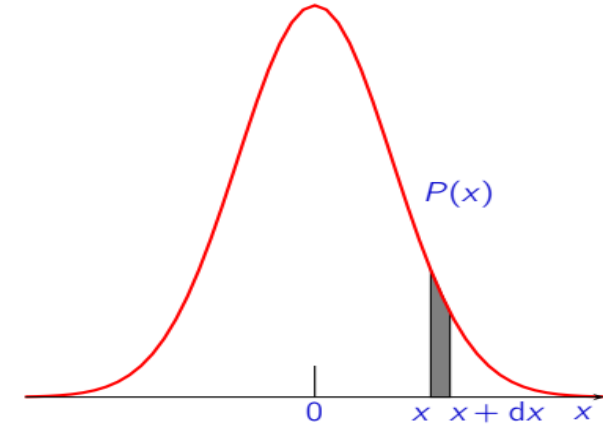
Example: We can approximate the vibrational motion of a diatomic molecule by a *harmonic oscillator* with effective mass m , force constant k , and position x . For the lowest-energy state the wavefunction is

$$\psi_0 = N \exp\left(-\frac{\sqrt{km}}{2\hbar} x^2\right)$$

where $\hbar$ is Planck's constant h divided by 2π , and N is a constant. This wavefunction is real, but **in general wavefunctions may be complex.**

For the harmonic oscillator ground-state wavefunction

$$\begin{aligned} P(x)dx &\propto \psi_0^* \psi_0 dx \\ &= N^2 \exp\left(-\frac{\sqrt{km}}{\hbar} x^2\right) dx \end{aligned}$$



for the probability of finding the particle between x and $x + dx$.

$P(x)$ is called the *probability density*

we must have $\int_{-\infty}^{\infty} P(x)dx = 1$. To achieve this, we need

$$P(x) = \frac{\psi_0^* \psi_0}{\int_{-\infty}^{\infty} \psi_0^* \psi_0 dx} \quad (2)$$

This expression is simplified if $\int_{-\infty}^{\infty} \psi_0^* \psi_0 dx = 1$. If this is true, the wavefunction is said to be *normalised*. We can achieve this by a suitable choice of the constant N . However N is arbitrary — it cancels out of the expression for $P(x)$ — so normalisation is a convenience, not a necessity. For ψ_0 (eq. (1)), the normalisation condition is

$$\int_{-\infty}^{\infty} N^2 \exp\left(-\frac{\sqrt{km}}{\hbar} x^2\right) dx = 1 \quad (3)$$

We get $N^2 \sqrt{\frac{\pi \hbar}{\sqrt{km}}} = 1$, and $N = \left[\frac{\sqrt{km}}{\pi \hbar}\right]^{1/4}$

$$\psi_0 = N \exp\left(-\frac{\sqrt{km}}{2\hbar} x^2\right)$$

波函数 ψ_0 是一个与时间无关的波函数：它不依赖于时间 t ，也不依赖于任何可以用它来预测的物理性质。在化学中使用的大多数波函数同样与时间无关。如果我们测量位置，我们就不能准确地预测结果。波函数 ψ_0 只通过概率密度 $P(x)$ 告诉我们找到任何特定值 x 的概率。

1.5 Operators and measures

波函数是概率波，它不代表任何真实物理量，但包含微观体系物理性质的信息。要预测系统的一个性质，就必须将一个算符作用在该波函数上。每一个可观测的物理量，如位置、能量或动量，都由一个算符来描述。

An operator is a mathematical instruction that converts one function into another

The simplest operator is the operator $\hat{x}$ which means ‘multiply the wavefunction by x ’. This is the operator we have just been using to calculate the expectation value of the position and its uncertainty. A similar operator which is almost as simple is the potential energy operator $\hat{V}$, which means ‘multiply by the function $V(x)$ ’, i.e.

$$\hat{V}\psi(x) = V(x)\psi(x)$$

If the system is a harmonic oscillator, then $V(x) = \frac{1}{2}kx^2$, in which case

$$\hat{V}\psi(x) = \frac{1}{2}kx^2\psi(x)$$

Differential operators

Another type of operator is *a differential operator*.

For instance, the operator $\frac{d}{dx}$ changes the function $\sin x$ into the function $\cos x$:

$$\frac{d}{dx} \sin x = \cos x$$

If a particle's position is described by coordinates (x,y,z) then its momentum in the x direction is described by the operator

$$\hat{p}_x = -i\hbar \frac{\partial}{\partial x}$$

(We have to use partial derivatives here because there are several coordinates.)

This equation is one of the fundamental postulates of quantum mechanics. It is by now universally accepted to be true because it leads to results that agree with experiment, with no measurable discrepancies since it was first proposed.

Constructing operators

We almost always work with momentum in quantum mechanics rather than velocity.

The rules for constructing the operator corresponding to a physical quantity are as follows:

- Write the expression for the physical quantity in terms of position and momentum variables.
- All position variables remain unchanged.
- A momentum in the direction of the coordinate q is replaced by the operator $-i\hbar\partial/\partial q$.

The last rule applies to angular coordinates as well as Cartesian ones.

For example, rotation about the z axis can be described using the spherical polar coordinate

φ , and then the angular momentum about the z axis is described by the operator $-i\hbar\frac{\partial}{\partial\varphi}$.

Mechanical quantities and their Operators

Mechanical quantities	Mathematical Operator
X	$\hat{X} = X$
P_x	$\hat{p} = -\frac{ih}{2\pi} \frac{\partial}{\partial x} = -i\hbar \frac{\partial}{\partial x}$
$T = p^2/2m$	$\hat{T} = -\frac{h^2}{8\pi^2 m} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) = -\frac{h^2}{8\pi^2 m} \nabla^2$
V	$\hat{V} = V$
$E = T + V$	$\hat{H} = -\frac{h^2}{8\pi^2 m} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) + \hat{V}$

Measurement

When a system is described by a wavefunction, the mean value of the observable in a series of measurements is equal to the expectation value of the corresponding operator

In quantum mechanics the expectation value of a quantity Q is calculated by the formula

$$\langle Q \rangle = \frac{\int \psi^* \hat{Q} \psi d\tau}{\int \psi^* \psi d\tau}$$

where $\hat{Q}$ is the operator corresponding to Q , and $\int ... d\tau$ is a conventional notation meaning that we integrate over *all variables* and over *all space*.

分子中的因子顺序很重要：算符只向右操作，因此在这种情况下 Q 仅作用于 ψ ，而不作用于 ψ^* 。这对于微分算符来说则至关重要。

Average Values

0, 20, 20, 60, 60, 80, 80, 80, 100

$$\frac{1}{N} = \frac{1}{9} \sum_j b_j = \frac{0 + 20 + 20 + 60 + 60 + 80 + 80 + 80 + 100}{9} = 56$$

$$\frac{1}{N} = \frac{1}{9} \sum_b n_b b = \frac{1(0) + 2(20) + 2(60) + 3(80) + 1(100)}{9} = 56$$

$$\langle B \rangle = \sum_{j=1}^N \frac{b_j}{N} = \sum_b \frac{n_b b}{N} = \sum_b \left(\frac{n_b}{N} \right) b = \sum_b P_b b$$

$$\langle x \rangle = \int_{-\infty}^{\infty} x |Y(x, t)|^2 dx$$

$$\langle x \rangle = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x |Y(x, y, z, t)|^2 dx dy dz$$

在接下来的许多章节中,我们将处理概率分布、平均值和标准差。因此,这里讨论概率的一些基本思想,并说明如何一般地计算平均值。

考虑一些如掷硬币或掷骰子的实验,其结果包含 n 种可能的结果,每种结果的概率为 $p_j (j=1, 2, \dots, n)$ 。如果实验重复无限次,则可凭直觉预测:

$$p_j = \lim_{N \rightarrow \infty} \frac{N_j}{N} \quad j=1, 2, \dots, n \quad (\text{B.1})$$

式中 N_j 是结果 j 出现的次数; N 是实验的总重复次数。因为 $0 \leq N_j \leq N$, 故 p_j 必须满足条件:

$$0 \leq p_j \leq 1 \quad (\text{B.2})$$

当 $p_j=1$ 时,我们认为事件 j 是必然的,而当 $p_j=0$ 时,我们认为这是不可能发生的。此外,由于

$$\sum_{j=1}^n N_j = N$$

可得归一化条件:

$$\sum_{j=1}^n p_j = 1 \quad (\text{B.3})$$

式(B.3)表示某个事件发生的概率是确定的。现在假设某个数字 x_j 与结果 j 是关联的,那么定义 x 的平均值为

$$\langle x \rangle = \sum_{j=1}^n x_j p_j = \sum_{j=1}^n x_j p(x_j) \quad (\text{B.4})$$

其中最后一项使用了扩展符号 $p(x_j)$, 表示得到结果 x_j 的概率。角括号 $\langle \rangle$ 表示一个物理量的平均值。

» 例题 B-1 假设有以下数据:

x	$p(x)$
1	0.20
3	0.25
4	0.55

计算 x 的平均值。

» 解 利用式(B.4), 可得

$$\langle x \rangle = (1)(0.20) + (3)(0.25) + (4)(0.55) = 3.15$$

将像 p_j 这样的概率分布解释为沿着 x 轴的单位质量的离散分布, 这样 p_j 是位于点 x_j 的质量分数。这种解释如图 B.1 所示。根据这一解释, x 的平均值是该系统的质心。

另一个重要物理量是

$$\langle x^2 \rangle = \sum_{j=1}^n x_j^2 p_j \quad (\text{B.5})$$

$\langle x^2 \rangle$ 称为 p_j 分布的二阶矩(second moment), 与转动惯量相似。

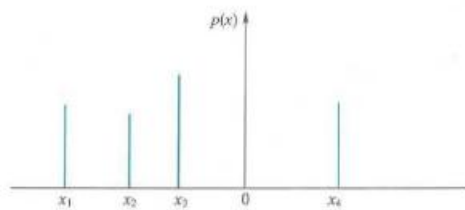


图 B.1 离散概率频率函数或概率密度 $p(x)$ 。

» 例题 B-2 计算例题 B-1 中所给数据的二阶矩。

» 解 利用式(B.5), 可得

$$\langle x^2 \rangle = (1)^2(0.20) + (3)^2(0.25) + (4)^2(0.55) = 11.25$$

从例题 B-1 和例题 B-2 可知, $\langle x \rangle^2 \neq \langle x^2 \rangle$ 。这个不等式是一个通式, 将在下文中证明。

在物理上比 $\langle x^2 \rangle$ 更有趣的物理量是二阶中心矩(second central moment)或方差(variance), 其定义为

$$\sigma_x^2 = \langle (x - \langle x \rangle)^2 \rangle = \sum_{j=1}^n (x_j - \langle x \rangle)^2 p_j \quad (\text{B.6})$$

如上所示, 式(B.6)中方差的平方根表示为 σ_x , 称为标准差(standard deviation)。从式(B.6)的求和中可以看出, σ_x^2 在 x_j 与 $\langle x \rangle$ 偏差较大时具有较大的数值, 因为在这种情况下, $(x_j - \langle x \rangle)$ 和 $(x_j - \langle x \rangle)^2$ 的值相对于 p_j 将很大。同时, 如果

Expectation values for general operators

Notice that $\langle Q \rangle$ is the same for the function $c\psi$ as it is for ψ , where c is any number (possibly complex). ψ and $c\psi$ are the same as far as any physical predictions are concerned. As we have seen, it is usually convenient to normalize the wavefunction, i.e. to choose c so that $\int \psi^* \psi d\tau = 1$. For a normalized wavefunction,

$$\langle Q \rangle = \int \psi^* \hat{Q} \psi d\tau$$

波函数是表达具有统计意义的概率波



微观粒子性质的表现具有统计性



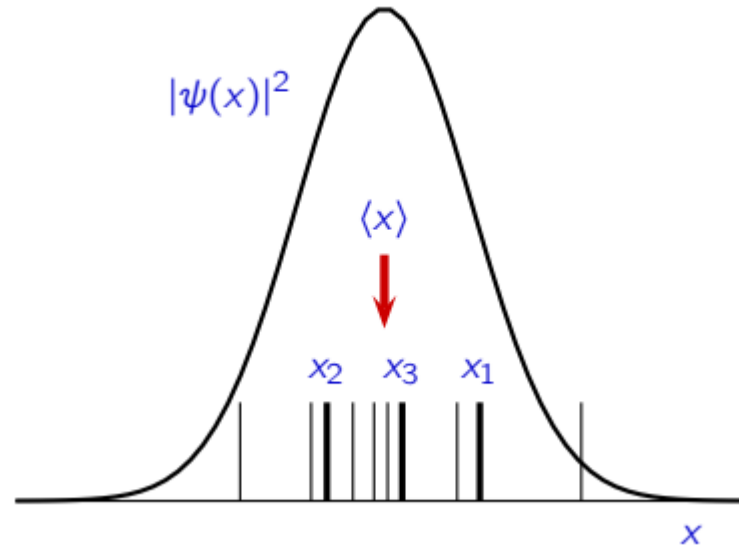
采用统计的方法计算微观粒子的力学量



力学量的计算值应为统计平均值



统计分布函数



The average or *expectation value* $\langle x \rangle$ of a large number of measurements of x is

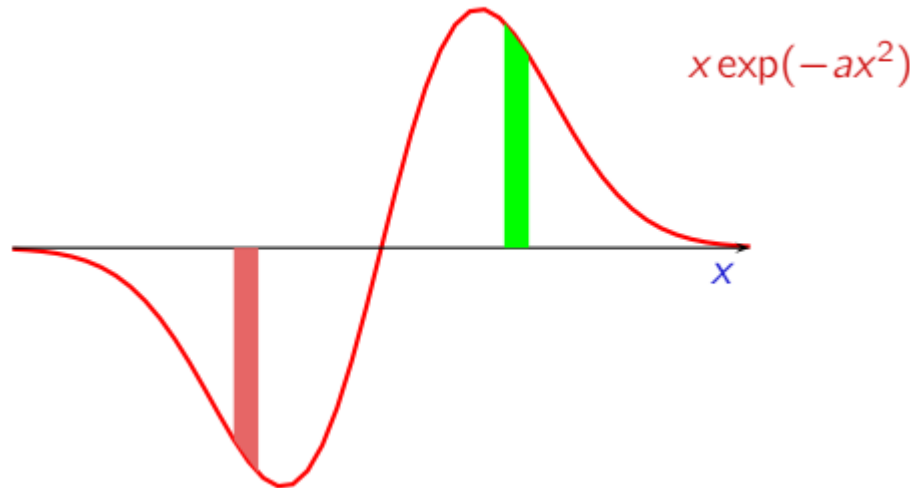
$$\langle x \rangle = \int_{-\infty}^{\infty} x P(x) dx = \int_{-\infty}^{\infty} \psi_0^*(x) x \psi_0(x) dx = \frac{\int_{-\infty}^{\infty} \psi_0^*(x) x \psi_0(x) dx}{\int_{-\infty}^{\infty} \psi_0^*(x) \psi_0(x) dx} \quad (4)$$

The mean position of an oscillator described by the wavefunction (1) is

$$\langle x \rangle = \int_{-\infty}^{\infty} xP(x)dx = N^2 \int_{-\infty}^{\infty} x \exp\left(-\frac{\sqrt{km}}{\hbar} x^2\right) dx = 0$$

using the fact that $\int_{-\infty}^{\infty} x \exp(-ax^2) dx = 0$.

This is fairly obvious from the fact that the probability distribution is symmetrical about $x = 0$ (an even function of x) so that the integrand is *odd*.



Uncertainty

Uncertainty:

$$\Delta Q = \sqrt{\langle \hat{Q}^2 \rangle - \langle \hat{Q} \rangle^2}$$

The uncertainty Δx in x is defined by $(\Delta x)^2 = \langle |x - \langle x \rangle|^2 \rangle = \langle x^2 \rangle - \langle x \rangle^2$

For the wavefunction ψ_0 (oscillator), we know that $\langle x \rangle = 0$. For $\langle x^2 \rangle$ we have

$$\begin{aligned} \langle x^2 \rangle &= \int_{-\infty}^{\infty} x^2 P(x) dx = \frac{\int_{-\infty}^{\infty} \psi_0^*(x) x^2 \psi_0(x) dx}{\int_{-\infty}^{\infty} \psi_0^*(x) \psi_0(x) dx} = \frac{\int_{-\infty}^{\infty} x^2 \exp(-ax^2) dx}{\int_{-\infty}^{\infty} \exp(-ax^2) dx} \\ &= \frac{(1/2a) \sqrt{\pi/a}}{\sqrt{\pi/a}} = \frac{1}{2a} = \frac{\hbar}{2\sqrt{km}} \quad \text{with } a = \sqrt{km}/\hbar \end{aligned}$$

$$\Delta x = \left[\hbar^2 / 4km \right]^{1/4}$$

$$\Delta x = \left[\hbar^2 / 4km \right]^{1/4}$$

- For mass 100g and force constant 6 N m⁻¹, $\Delta x \approx 8 \times 10^{-8} \text{Å}$.

This system would oscillate at about 1 Hz.

- For a diatomic molecule with (reduced) mass 1a.u and force constant 600 N m⁻¹, $\Delta x \approx 0.07 \text{Å}$.

Notice that the uncertainty in the position of the macroscopic object is much smaller in absolute terms, not just in relative terms.

The Heisenberg uncertainty principle

Uncertainty is a fundamental aspect of quantum mechanics, expressed in its simplest form by Heisenberg's uncertainty principle for position and momentum.

If x is a position coordinate and p_x is the corresponding momentum (formally, the 'momentum conjugate to x ', in the jargon), then the uncertainties in x and p_x satisfy the inequality

$$\Delta x \Delta p_x \geq \frac{1}{2} \hbar$$

Note that this relationship has nothing to do with experimental error; it is a fundamental feature of the theory. It tells us that, even in an ideal experiment, it is *impossible* to measure both the momentum and the position of a particle precisely.

- 经典: 测量中的误差取决于仪器的精度, 可以任意小.
- 量子: 在物理上, 不可能同时测量一个粒子的精确位置和精确速度。

Example

The speed of an electron is measured to be 1000 m/s to an accuracy of 0.001%. Find the uncertainty in the position of this electron.

$$p = mv = (9.11 \times 10^{-31} \text{ kg}) (1 \times 10^3 \text{ m/s}) = 9.11 \times 10^{-28} \text{ kg.m/s}$$

$$\Delta p = p \times 0.001\% = 9.11 \times 10^{-33} \text{ kg m/s}$$

$$\Delta x = h/\Delta p = 6.626 \times 10^{-34} / (9.11 \times 10^{-33}) = 7.27 \times 10^{-2} \text{ (m)}$$

The speed of a bullet of mass of 0.01 kg is measured to be 1000 m/s to an accuracy of 0.001%. Find the uncertainty in the position of this bullet.

$$p = mv = (0.01 \text{ kg}) (1 \times 10^3 \text{ m/s}) = 10 \text{ kg.m/s}$$

$$\Delta p = p \times 0.001\% = 1 \times 10^{-4} \text{ kg m/s}$$

$$\Delta x = h/\Delta p = 6.626 \times 10^{-34} / (1 \times 10^{-4}) = 6.626 \times 10^{-30} \text{ (m)}$$

1.6 The time-independent Schrödinger equation

For which operators does a wave function satisfy an eigenvalue equation? In general, this is not an easy question to answer and depends on the system. For the total energy operator, which is called the *Hamiltonian*, $\hat{H}$, the resulting eigenvalue equation is

$$\hat{H}\psi = E\psi$$

which is the *time-independent Schrödinger equation*. This is the central equation of quantum chemistry, because solving it allows us to calculate ψ . Because ψ is an eigenfunction of $\hat{H}$, we know that an experiment can measure the energy E precisely, with zero uncertainty. $\hat{H}$ is the sum of the kinetic and potential energy operators $\hat{T}$ and $\hat{V}$:

$$\hat{H} = \hat{T} + \hat{V}$$

We can not derive or prove Schrödinger equation

Example: the harmonic oscillator Hamiltonian

As an example we can construct the Hamiltonian for the harmonic oscillator.

The kinetic energy $T = \frac{1}{2}mv^2$ needs to be written in terms of the momentum $p_x = mv$, i.e. as $p_x^2/2m$. The potential energy for a harmonic oscillator is $V = \frac{1}{2}kx^2$. Replacing x and p_x by their operators gives

$$\hat{T} = \frac{1}{2m} \hat{p}_x^2$$

$$\hat{V} = \frac{1}{2} k \hat{x}^2$$

and therefore

$$\hat{H} = \frac{1}{2m} \hat{p}_x^2 + \frac{1}{2} k \hat{x}^2$$

Substituting $\hat{x} = x$ and $\hat{p}_x = -i\hbar d/dx$, yields $\hat{H} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + \frac{1}{2} kx^2$

Hence the Schrödinger equation of a harmonic oscillator is $-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \psi + \frac{1}{2} kx^2 \psi = E\psi$

Time-dependent wavefunctions

So far we have discussed time-independent wavefunctions ψ , which yield observables that are independent of time. Almost all of the wavefunctions you will encounter in chemistry are of this type, but it is useful to know that there exists a more general type of *time-dependent* wave function, which we will write $\Psi(x, t)$. These functions satisfy the *time-dependent* version of the Schrödinger equation:

$$\hat{H}\Psi(x, t) = i\hbar \frac{\partial \Psi}{\partial t} \quad (*)$$

If the Hamiltonian is independent of time, then there exists a special class of solutions of the form,

$$\Psi(x, t) = \psi(x) e^{-iEt/\hbar}$$

where $\psi(x)$ is the time-independent wave function.

$$\hat{H}\Psi(x,t) = i\hbar \frac{\partial \Psi}{\partial t}$$

$$\frac{\partial \Psi}{\partial t} = -\frac{i}{\hbar} \hat{H}\Psi$$

- (1) 只有粒子状态量，没有运动量，描述的是粒子状态演化规律，无法提供中间状态的信息
- (2) 方程有关于时间的一阶导数，需要与时间有关的边界条件
- (3) 全空间找到粒子的概率是守恒的（板书）

1.7 The time-dependent Schrödinger equation

If $V(x,t) = V(x)$

$$-\frac{\hbar}{i} \frac{\partial \Psi(x,t)}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \Psi(x,t)}{\partial x^2} + V(x) \Psi(x,t)$$

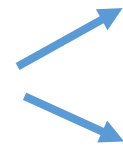
$$\Psi(x,t) = f(t)\psi(x)$$



$$\begin{aligned} \frac{\partial \Psi(x,t)}{\partial t} &= \frac{df(t)}{dt} \psi(x) \\ \frac{\partial^2 \Psi(x,t)}{\partial x^2} &= f(t) \frac{d^2 \psi(x)}{dx^2} \end{aligned}$$

$$-\frac{\hbar}{i} \frac{df(t)}{dt} \psi(x) = -\frac{\hbar^2}{2m} f(t) \frac{d^2 \psi(x)}{dx^2} + V(x) f(t) \psi(x)$$

$$-\frac{\hbar}{i} \frac{1}{f(t)} \frac{df(t)}{dt} \psi(x) = -\frac{\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2} + V(x)\psi(x)$$



$$-\frac{\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x)$$

$$\frac{df(t)}{f(t)} = -\frac{iE}{\hbar} dt \quad f(t) = e^{-iEt/\hbar}$$

$$\Psi(x, t) = e^{-iEt/\hbar} \psi(x)$$

Stationary states

For the wavefunction $\Psi = \psi(x)e^{-iEt/\hbar}$ the expectation value of any operator which doesn't itself depend on the time takes the form

$$\langle Q \rangle = \frac{\int \Psi^* \hat{Q} \Psi d\tau}{\int \Psi^* \Psi d\tau} = \frac{\int (\psi e^{-iEt/\hbar})^* \hat{Q} (\psi e^{-iEt/\hbar}) d\tau}{\int (\psi e^{-iEt/\hbar})^* (\psi e^{-iEt/\hbar}) d\tau} = \frac{\int \psi^* \hat{Q} \psi d\tau}{\int \psi^* \psi d\tau}$$

波函数的时间依赖性抵消，期望值与时间无关。这种波函数称为定态，这些是在化学中通常遇到的状态。我们通常忽略 $\exp(-iEt/\hbar)$ 中包含的时间依赖性，而只考虑与时间无关的波函数 $\psi(x)$ ，它满足与时间无关的薛定谔方程。有更一般的时间相关的波函数，其中观测确实依赖于时间。

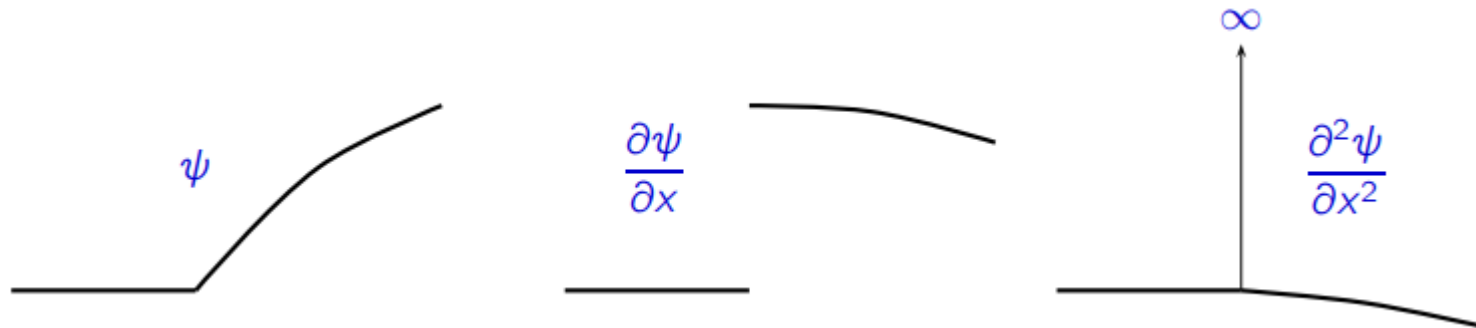
Continuity

The kinetic energy T involves the square of the momentum, which becomes a second derivative in operator form. The wavefunction has to satisfy the Schrödinger equation

$$\hat{H}\psi = \hat{T}\psi + \hat{V}\psi = E\psi$$

ψ remains finite (remember that it describes the probability density) so if V is finite then $\hat{V}\psi$ is finite. Therefore $\hat{T}\psi$ must remain finite. This means that ψ and its first derivative must be *continuous*.

If V becomes infinite, and ψ is nonzero, then it is possible (in fact necessary) for $\hat{T}\psi$ to be infinite also, which means that there is a discontinuity in the derivative of ψ . However ψ itself must still b



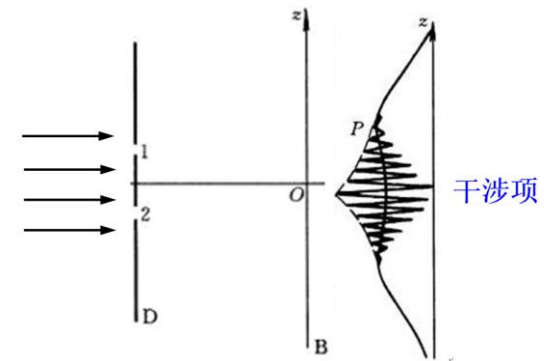
1.8 Discussions for the Schrödinger equation

(a) Superposition Principle

Mathematically, superposition principle means, if you linearly combine any number of solutions of an equation then the linear combination is itself a solution. The basic equation of quantum mechanics is **Schrödinger equation, which is a linear differential equation**, so its solutions obey superposition principle

If $\varphi_1, \varphi_2, \dots, \varphi_n$ are the possible states of a microscopic system, then the linear combination of these states is also a possible state of the system.

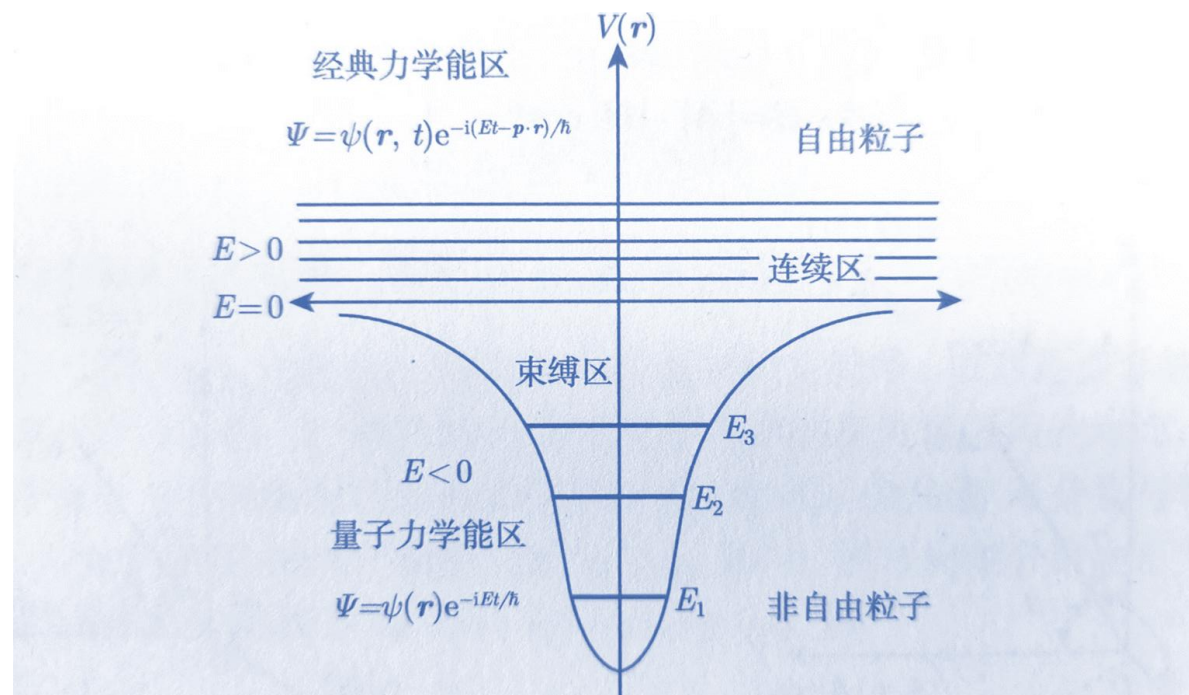
$$\Psi = \sum_n c_n \varphi_n$$



(b) The emergence of quantization

薛定谔方程是一个微分方程，有无穷解。对于E的任何值，它都有数学上可接受的解。然而，Born的波函数概率解释对此有限制：当系统的边界将粒子限制在一个有限的区域时，几乎所有的解都是不可接受的，要求波函数连续且在边界处为特定的解；可接受的解只出现在特殊值下 → 能量量子化是边界条件的结果。

$$-\frac{\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x)$$
$$-\frac{\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2} = (E - V(x))\psi(x)$$



粒子所处的状态可分为**束缚态**（**bound state**）和**非束缚态**。如果满足条件

$$\lim_{x \rightarrow \pm\infty} \Psi(x) \rightarrow 0$$

该状态称为**束缚态**。否则称为**非束缚态**。处于束缚态的粒子被束缚在空间一定范围内。非束缚态又称为**游离态**、**自由态**。

定义量子力学中的**束缚态**和**非束缚态**来说就要考虑边界条件了，那就是无限远的势能。只要粒子在无限远处，不管是那个方向的无限远处能量低于势能，它就处于束缚态。如果粒子在无限远处，只要有一个方向粒子能量高于势能粒子就处于散射态。

Free particles

Classically, a particle moving in one dimension without external forces has constant momentum. Quantum mechanically we expect to be able to find states of definite momentum. If the motion is along the x axis we need to look for solutions of the eigenvalue equation

That is,

$$\hat{p}_x \psi(x) = p \psi(x)$$

$$-i\hbar \frac{d}{dx} \psi = p \psi$$

The solution to this equation is

$$\psi_p = \exp(ipx / \hbar)$$

Remember that p is an eigenvalue — a *constant* with (in this case) dimensions of momentum.

The time-independent Schrödinger equation is

$$H\psi = -\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} = E\psi$$

and the general solution of this is

$$\psi = ae^{ikx} + be^{-ikx}$$

where $k = \sqrt{2mE}/\hbar$. If $E = p^2/2m$, then $k = p/\hbar$ and we arrive at the result

$$\psi = a\psi_p + b\psi_{-p}$$

This ***superposition*** or ***linear combination*** of two wavefunctions, both with energy $p^2/2m$, is also an eigenfunction of **H** with energy $p^2/2m$, for any values of the constants a and b .

$$\psi = A\sin(px/\hbar) + B\cos(px/\hbar)$$

However it is not an eigenfunction of the operator p_x , unless $a = 0$ or $b = 0$, so it doesn't have a definite momentum. We can write the same wavefunction in the form

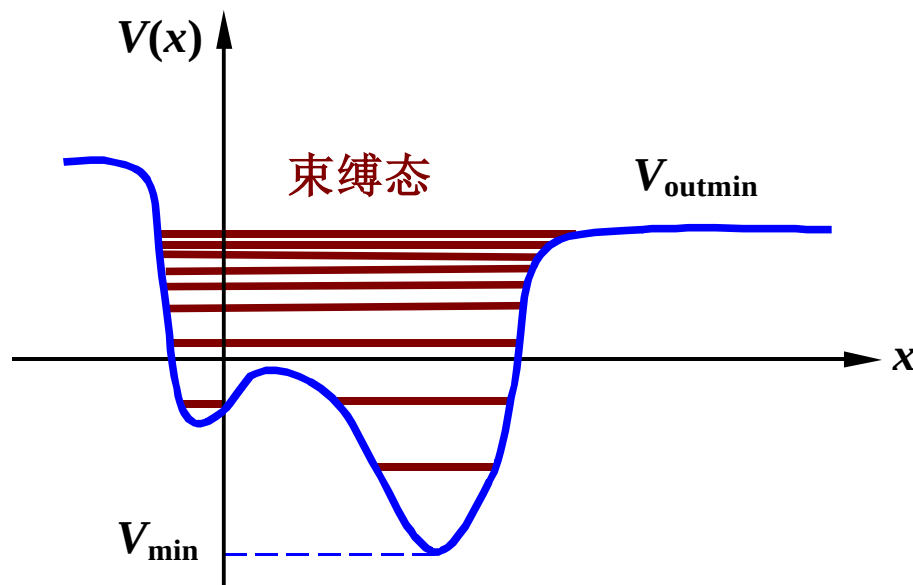
$$\psi = A \sin(px/\hbar) + B \cos(px/\hbar)$$

$$\int_{-\infty}^{\infty} \psi^*(x) \psi(x) dx \quad \text{infinite}$$

The free-particle problem is an **unreal** situation because we could not actually have a particle that had no interaction with any other particle in the universe.

束缚态的性质决定了物质的性质，如原子和分子的结构、光谱等。例如，化学体系的电子是通过束缚态来描述的，不同的束缚态对应了不同的轨道和能级。这也解释了为什么原子只能吸收或放出特定的能量光子，而不能取任意能量的原因。散射态是指粒子在势能场中穿越或被势能场散射的状态。在散射态下，粒子的运动范围不再受限制，可以自由传播。与束缚态不同，散射态的能量是连续的，可以取任意值。

束缚态的能量满足条件 $V_{\min} < E < V_{\text{out min}}$



一维束缚态的非简并性

考虑一维两个束缚态 $\psi_a(x)$ 和 $\psi_b(x)$ ，如果 $E_a=E_b$ ，那么这两个波函数描述同一个状态

$$\frac{2m}{\hbar}(E_a - V)\psi_a = -\frac{d^2\psi_a}{dx^2} \quad (*)$$

$$\frac{2m}{\hbar}(E_b - V)\psi_b = -\frac{d^2\psi_b}{dx^2} \quad (**)$$

$$\psi_b \times (*) - \psi_a \times (**)$$

$$\frac{2m}{\hbar}(E_a - E_b)\psi_a\psi_b = \frac{d^2\psi_b}{dx^2}\psi_a - \frac{d^2\psi_a}{dx^2}\psi_b = \frac{d}{dx}\left(\frac{d\psi_b}{dx}\psi_a - \frac{d\psi_a}{dx}\psi_b\right)$$

因为 $E_a = E_b$ ，上式左边为零

$$\frac{d}{dx}\left(\frac{d\psi_b}{dx}\psi_a - \frac{d\psi_a}{dx}\psi_b\right) = 0$$

$$\left(\frac{d\Psi_b}{dx}\Psi_a - \frac{d\Psi_a}{dx}\Psi_b\right) = C \quad (\text{常数})$$

因为

$$\lim_{x \rightarrow \pm\infty} \Psi(x) \rightarrow 0$$

$$\left(\frac{d\Psi_b}{dx}\Psi_a - \frac{d\Psi_a}{dx}\Psi_b\right) = 0$$

上式左右两边同时除以 $\Psi_a \Psi_b$

$$\frac{\frac{d\Psi_b}{dx}\Psi_a - \frac{d\Psi_a}{dx}\Psi_b}{\Psi_a \Psi_b} = 0$$

$$\frac{d}{dx} \left[\ln \left(\frac{\Psi_b}{\Psi_a} \right) \right] = 0 \quad \Rightarrow \quad \ln \left(\frac{\Psi_b}{\Psi_a} \right) = \text{constant}$$

$$\Psi_b = A \Psi_a$$

理论框架与数学基础

物理图像

旧量子论：“经典力学+量子化条件

粒子的图像

新量子论： 矩阵力学（海森堡，1925）
波动力学（薛定谔，1926）

波粒二象性